KisanConnect

Intelligent Crop Recommendation, Disease Detection & Yield Prediction

Theme: AI in Agriculture (AgriTech)

TEAM NAME

Code Hunters

TEAM MEMBER NAMES

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PROBLEM STATEMENT

Farmers still decide on guesswork - there is no single smart, accessible tool for crop decisions.

Generic Advice

Crop and fertilizer tips ignore soil type, climate and nutrient levels, so yields stay below potential.

Late Disease Detection

Manual inspection is slow. Diseases spread and cause heavy crop loss before action is taken.

Unreliable Yield Forecasts

Without accurate prediction, farmers cannot plan resources, crop cycles or market sales.

Rural Access Gap

Remote farmers lack modern tools, expert advice and timely, language-friendly insights.

Result: lower yields, wasted inputs and avoidable losses for farmers.

PROPOSED SOLUTION

One AI platform: soil data or a leaf photo in, clear decisions out.

99.3%	Rule-first	38	1.10
Crop Recommendation	**Fertilizer Advice**	**Disease Classes**	**Yield Prediction (MAE)**
Random Forest, hold-out	Soil test + crop need	CNN on PlantVillage photos	t/ha, time-split, no leakage

Farmer signs up (admin verified) → **Enters soil data / uploads leaf photo** → **AI models analyse** → **Crop, fertilizer, cure, yield + report**

2 Languages
Hindi + English now; Tamil, Telugu, Punjabi, Kannada soon

Auto Reports
Profit and revenue per acre, downloadable

AI Chatbot
24/7 farming help in simple language

Supplement Store
Cures and fertilizers with buy links

DATA & EVALUATION

Every number below is labelled with its dataset and test method.

Crop Recommendation

99.3% accuracy (80/20 hold-out)

Data: Kaggle crop dataset, 22 crops; Random Forest.
Note: Near-separable classes, so not a field-accuracy claim.

Fertilizer Advice

Rule-based, RF only as a hint

Data: Kaggle table of 99 rows; rule from soil test + crop nutrient need.
Note: Only 99 rows, so a high RF score is memorisation.

Disease Detection

97.82% (reported by base project)

Data: PlantVillage, 54,000+ leaf images, 38 classes; CNN.
Note: Lab-style photos: accuracy on real field photos will be lower.

Yield Prediction

MAE 1.10 t/ha, R^2 0.66

Data: Kaggle Indian crop yield; train to 2016, test 2017-2020.
Note: No Production input (no leakage); 5-year median was best.

Metrics come from the KrishiDisha report and model cards (open-source base project).

TECH STACK

STACK

Frontend	HTML5, CSS3, JavaScript (responsive, multilingual)
Backend	Python, Flask REST API, SQLAlchemy, MySQL
AI / ML	Scikit-learn (RF, GBR), TensorFlow/Keras (CNN), OpenCV, NLP chatbot
Data & Tools	Kaggle datasets (PlantVillage), Colab, Git/GitHub, VS Code

ARCH. DIAGRAM

Farmer | Admin (verifies farmers)

⌄

Web UI: registration, dashboard, image upload

⌄

Flask REST API: validate, preprocess, route

⌄

Random Forest
Crop + Fertilizer

CNN
Disease

Gradient Boost
Yield

⌄

MySQL (users, activity, results) + image storage

BUSINESS PLAN

Target Users

- Small and marginal farmers
- Farmer groups and agri-cooperatives
- Agri institutions and NGOs

Revenue Streams

- Commission on supplement and fertilizer sales
- Premium reports and financial insights
- Licensing to institutions and NGOs

Go-to-Market

- Pilot with local farmers
- Partner with agri institutions
- Hindi-first, mobile-friendly rollout

Growth Roadmap

- Mobile app with offline mode
- IoT sensors and live market prices
- More crops, diseases and languages

Farmers save on wasted inputs and losses; we earn when they buy and subscribe.

THANK YOU

Team Code Hunters

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